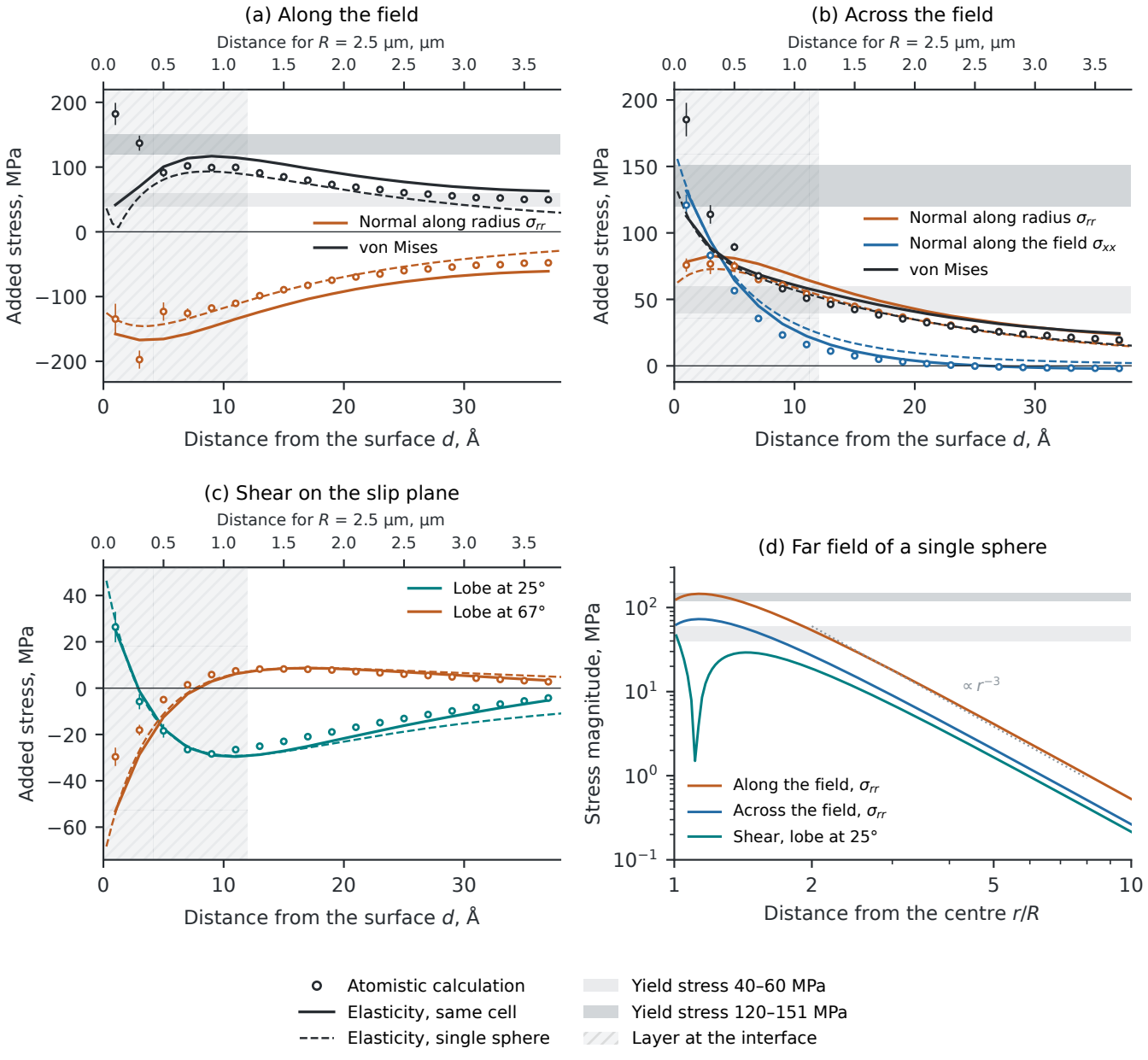
(a) Along the field
Distance for R = 2.5 μm, μm
0.0 0.5 1.0 1.5 2.0 2.5 3.0 3.5
Added stress, MPa
200 100 0 −100 −200
Normal along radius σ_{rr}
von Mises
0 10 20 30
Distance from the surface d, Å
(b) Across the field
Distance for R = 2.5 μm, μm
0.0 0.5 1.0 1.5 2.0 2.5 3.0 3.5
Added stress, MPa
200 150 100 50 0
Normal along radius σ_{rr}
Normal along the field σ_{xx}
von Mises
0 10 20 30
Distance from the surface d, Å
(c) Shear on the slip plane
Distance for R = 2.5 μm, μm
0.0 0.5 1.0 1.5 2.0 2.5 3.0 3.5
Added stress, MPa
40 20 0 −20 −40 −60
Lobe at 25°
Lobe at 67°
0 10 20 30
Distance from the surface d, Å
(d) Far field of a single sphere
Stress magnitude, MPa
10^2 10^1 10^0 10^{-1}
$\propto r^{-3}$
Along the field, σ_{rr}
Across the field, σ_{rr}
Shear, lobe at 25°
1 2 5 10
Distance from the centre r/R
Atomistic calculation
Elasticity, same cell
Elasticity, single sphere
Yield stress 40–60 MPa
Yield stress 120–151 MPa
Layer at the interface